

Sri Datta Ganesh Bandreddi

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EDUCATION

CARNEGIE MELLON UNIVERSITY

Master of Science in AI Engineering, Biomedical Engineering | GPA: 4.00/4.00

Pittsburgh, PA

Dec 2026

VISVESVARAYA NATIONAL INSTITUTE OF TECHNOLOGY

Bachelor of Technology, Mechanical Engineering | GPA: 8.44/10.00

Nagpur, India

Jul 2022

SKILLS

Languages: Python, C++, Go, Java, SQL, Bash /Shell Scripting

GPU (Acceleration): CUDA, cuDNN, cuBLAS, TensorRT, Triton Inference Server, NCCL, Nsight Systems/Compute

DL Frameworks: PyTorch, JAX, TensorFlow, HF, NeMo/Megatron-LM, LoRA/QLoRA, FP8/BF16 Mixed Precision

Data & ETL: Rapids (cuDF, cuML), NVTabular, DALI, NumPy, SciPy, Pandas, SparkML

Systems & MLOps: Docker, Kubernetes, Slurm, Jenkins/GitHub Actions, MLflow, Git, AWS (ECS, SageMaker), GCP

Data Infrastructure: PostgreSQL, Neo4j, Kafka, Redis, Iceberg, Elasticsearch, FAISS, MongoDB

PROFESSIONAL EXPERIENCE

MOTIONAL

Pittsburgh, PA

Data Platform Intern (Infrastructure)

May 2026 - Aug 2026

- Enabled teams to understand petabyte-scale drivelog data before thawing it, by designing and shipping a metadata-extraction system end to end (staging PoC through production design) on a medallion architecture
- Enabled fast scene-metadata retrieval and earned a provisional patent, by building a geospatial indexing scheme (H3) and scene embeddings for similarity search
- Shipped natural-language drivelog search and selected the production data stack, by building an agentic LangGraph layer on a FastAPI backend and benchmarking Ray Data against Spark on Kubernetes

PHENOM

Hyderabad, India

Product Development Engineer (Enterprise Talent Graph)

Dec 2023 - Jul 2025

- Improved scalability and reliability of a multi-tenant AI analytics platform under heavy concurrent load, by leading six engineers to refactor shared services and cut cross-tenant coupling
- Scaled a production knowledge graph past 1M entities with continuous ingestion, by designing validated aggregation pipelines on Neo4j schemas and Iceberg versioned storage
- Cut AI-driven query response time by 40% across 100+ clients, by tuning PostgreSQL sharding and hardening the Kafka platform (custom exporter, Kafka UI, ZooKeeper-to-KRaft migration)

BYJU'S

Hyderabad, India

Software Engineer (Core Platform Team)

Jul 2022 - Dec 2023

- Cut infrastructure costs by 70% for 1M+ users, by deploying scheduled AWS ECS auto-scaling and Terraform IaC tuned to real traffic patterns
- Halved reliance on external identity providers, by leading development of an in-house auth platform (SAML, SSO, RBAC) in Go and Java (Spring Boot)

PHENOM

Hyderabad, India

Machine Learning Intern

May 2021 - Jun 2022

- Boosted recommendation accuracy for production pipelines, by building persona-classification models with spaCy and BERT embeddings across 3 data sources
- Cut recommender preprocessing latency, by migrating CPU pipelines to NVTabular on GPU across 2 production databases

PROJECT EXPERIENCE

FLARE: Efficient Attention (Carnegie Mellon University) (link)

Aug 2025 - Present

- Co-authored a linear-attention operator (3rd of 7 authors, submitted to TMLR) later adopted by an NVIDIA and General Motors crash-dynamics effort for roughly 2x lower memory
- Demonstrated over 200x speedups on Long Range Arena (1k to 8k tokens), by benchmarking and profiling multi-GPU NCCL training across FP8/BF16 with Nsight Systems, and extended FLARE to image classification and diffusion

Distributed LLM Fine-Tuning with Megatron-LM (Carnegie Mellon University)

Aug 2025 - Oct 2025

- Fine-tuned a 7B-parameter LLM to stable convergence with about 40% less memory on four H100 GPUs, by configuring tensor and pipeline parallelism, gradient checkpointing, and LoRA/QLoRA adapters
- Raised training throughput to 2.1x over naive data parallelism, by tuning NCCL, gradient bucketing, and optimizer-state sharding

CT-Transformer, Vision Transformers (Carnegie Mellon University)

Sep 2025 - Dec 2025

- Reached 96.22% macro AUC-ROC on 15-class VinBigData chest X-ray classification (beating ResNet-50, EfficientNet-B3, and ViT baselines), by training a 210M-parameter Swin Transformer
- Cut inference latency by 3x versus PyTorch eager, by converting to TensorRT (FP16/INT8), accelerating preprocessing with DALI, and serving on Triton with adaptive batching